

V800

36 BAY · 4U RACK · V5 VIDEO SERVER

NDA

CE / UKCA

FCC / UL / RCM

THROUGHPUT

4,000 Mbit/s

MAX STORAGE

864 TB

DRIVE BAYS

36

MAX CAMERA STREAMS

325

RECOMMENDED USAGE

Designed for campus-wide deployments with centralized management, this solution delivers critical protection, high performance, and top-tier security. It supports high camera density with exceptional reliability, for robust, scalable surveillance across large environments.

KEY ATTRIBUTES

- Windows Server 2022 or 2025
- 32GB RAM Registered ECC
- RAID 60 data protection
- N+1 hot-swap power & cooling
- AMD EPYC 9005 Zen5 CPU
- Mirrored, hot-swap OS SSDs
- Quad 10Gb Ethernet
- TCG 2.0 cybersecurity w/ TPM

V800 front 3/4 — product photography



5-Year NBD Advanced Parts Warranty

5 Years, Next Business Day, Advanced Parts Replacement warranty ensures uptime continuity. Secure remote management & advanced support delivers rapid response & uptime.

VIDEOX ENTERPRISE FEATURES

- Lower deployment costs — hardware accelerated H.265**
V5 micro-architecture acceleration delivers up to 2.3× H.265 stream processing per server, and 30–50% storage and bandwidth savings versus H.264 — fewer servers, higher camera density, longer retention, same budget.

- High data availability**
RAID 6/60 redundancy tolerates up to 6 hard drive failures depending on model and configuration, with hardware XOR, CacheVault protection and patrol read repairs.

- Flexible & scalable tier-1 enterprise storage**
Scale to petabytes of tier-1 capacity and tailor retention per project. Enterprise-class HDDs and SSDs throughout, for tier-1 availability and performance.

- Strengthen cybersecurity**
A built-in TPM 2.0 module supports hardware Root of Trust authentication and data encryption. All VideoX are fully NDA compliant.

VALIDATED
WITH

Milestone

Genetec

Avigilon

Hanwha WAVE

NX Witness

Exacq

WHERE THE V800 SITS IN THE VIDEOX NVR LINE

Drive bays · max camera streams

V100
2 bay · 1U
25

V200
4 bay · 1U
100

V400
8 bay · 2U
200

V500
12 bay · 2U
275

V600
16 bay · 3U
275

V700
24 bay · 4U
325

V800
36 bay · 4U
325

MAXIMUM VIDEO STREAM RATE

4,000 Mbit/s · 864 TB raw · 720 TB usable

RESOLUTION	CODEC	CAMERA STREAMS	VS. 4MP BASELINE
4MP · 2560×1440 (16:9)	H.265-20	325	Baseline VSR
8MP · 3840×2160	H.265-20	179	~45% — 2× the pixels per frame

VSR PARAMETERS

RESOLUTION 4MP · 2560×1440

FRAME RATE 15 fps

CODEC H.265-20 (Good) · ~3.2 Mbit/s

RECORDING On motion, VMD + metadata

MOTION ACTIVITY 75% average per day

RETENTION 30 days

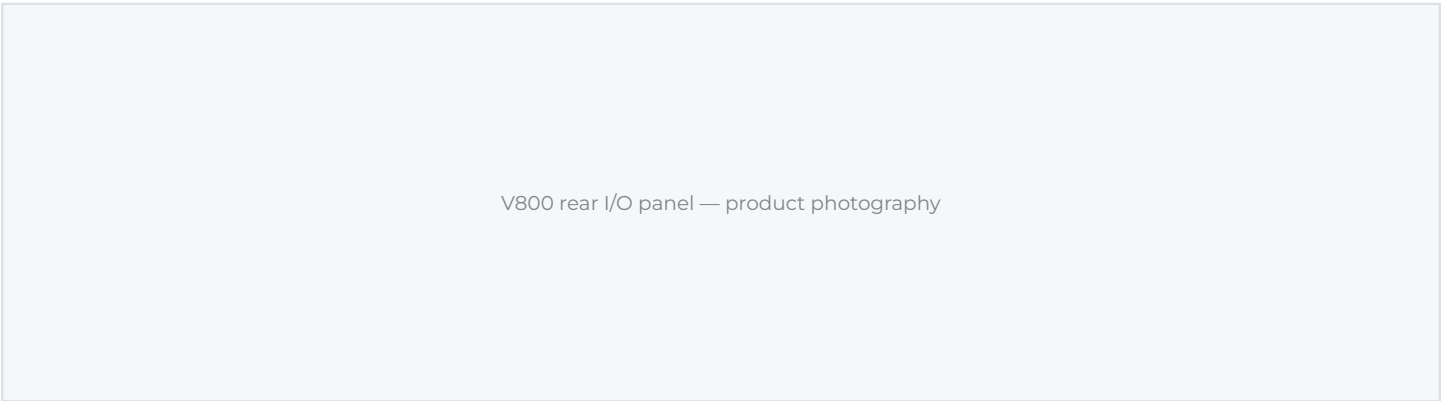
Max VSR is camera streams, not cameras — a multisensor device presents several streams. If a recording parameter changes, revise the calculation. Validated for Avigilon, Milestone, Hanwha/NXWitness and Genetec; other platforms vary.

ORDERABLE CONFIGURATIONS

PART NUMBER	DRIVE CONFIGURATION	RAW	USABLE · RAID 60
VX5-V800-576	36 × 16TB enterprise HDD	576 TB	480 TB
VX5-V800-720	36 × 20TB enterprise HDD	720 TB	600 TB
VX5-V800-864	36 × 24TB enterprise HDD	864 TB	720 TB

Three drive capacities per NVR — same chassis and performance, different retention. Usable capacity is RAID 60, excluding the mirrored OS SSD pair.

REAR I/O PANEL



GENERAL INFORMATION

Systems ship with Microsoft Windows pre-installed without media and not set up; the installer is responsible for setup, VMS installation, and configuration of users, cameras and databases. VMS installers are included but not installed. Milestone and Avigilon best practice is SQL Standard above 300 cameras — optional, not included. Client View application on server is not supported; a dedicated Client View workstation is required.

HARDWARE INFORMATION

CPU	5th Gen Zen5 AMD EPYC™ 9135, 4.3GHz max all-core turbo, 16C / 32T, 64MB cache, 200W TDP
Acceleration	Chiplet microarchitecture · AMD Infinity Architecture & Guard · CXL 2.0 · native 512-bit data paths · H.265 math acceleration
RAM	32GB ECC Registered DDR5 (minimum)
Operating system	Microsoft Windows Server Standard 2022 or 2025 (LTSC)
VMS / OS drive	Mirrored 480GB Enterprise 3D TLC flash SSD, dedicated for OS/VMS, hot-swap
Hard drives	Up to 36× Enterprise class 7200 RPM 3.5" HDD, 2.5M MTBF, 600k duty cycle, certified 24/7, tool-less hot-swap
Storage capacity	Up to 864TB raw capacity
RAID	Hardware RAID 6 double fault tolerance w/ HW XOR engine, CacheVault protection, patrol read repairs
Network	4× Enterprise 10Gb Eth RJ45 + 1Gb IPMI (2× 10Gb SFP+ or 2× 25Gb SFP28 upgrade available)
Display	VGA — Client View applications not supported on server
Encryption	TPM 2.0 (FIPS, CC-TCG certified) · Silicon Root of Trust · data-at-rest encryption with SEDs · AMD SEV/SME
Management	Out of Band (OOB) remote management: IPMI system management, KVM console redirection

REGULATORY & ENVIRONMENTAL

Form factor	Standard 19" rackmount w/ rails, 4U height
Power	1200W 1+1 redundant PSU, PMBus 1.2, 80+ Platinum · 100–127VAC 10A 50–60Hz (800W) · 200–240VAC 8A 50–60Hz (1200W)
Max draw	1200W, up to 80% efficient (Platinum), hot-plug redundant
Cooling	6× 80×38mm PWM low-power-consumption hot-swap fans
Dimensions	430mm (w) × 680mm (d) × 175mm (h)
Weight	Ship weight w/ 36 HDDs = 72kg / 167lbs
Operating temp	10–30°C / 41–86°F
Storage temp	–40–65°C / –40–149°F
Humidity	10–90% relative humidity, non-condensing
Safety	CE (class A), UKCA, FCC, RCM, UL
Trade	NDAA compliant, no disclosures